

Generative AI video models like MovieGAN and OpenAI Sora, which aim to generate realistic videos from text input, suffer from similar issues right now. If their training data doesn't capture the full range of human motion, it will generate human forms capable of performing physically impossible movements – as these AI generated videos of human gymnasts very well emphasise. Last year, a TikTok user self-released a song called “Heart On My Sleeve,” where the vocals sounded eerily similar to Drake and The Weeknd.

The song clocked over 15 million views on TikTok, not to mention hundreds of thousands more on Spotify and YouTube. If the viral hit was generated using an AI-based sound generation tool, chances are it might have been heavily trained on Western hip-hop music as part of its dataset, and that it won't be great at generating vocals that sound like Lata Mangeshkar. Probably why we haven't heard an AI generated song of a famous Indian singer yet, because of the lack of quality training data.

Could AI hallucinations also be linked to a lack of effort? Because the AI genie is well and truly out of the bottle, and there's no going back to non-GenAI times, companies and startups are locked in a furious race to release half-baked AI products to gain first mover's advantage and cover market share. These are some of the key findings of a recent report from Aporia, which surveyed about 1000 AI and ML professionals from North America and UK – individuals working in companies ranging from 500 to 7,000 employees, across various important sectors such as finance, insurance, healthcare and travel, among others.

Aporia's findings reveal a noteworthy trend among engineers working with LLMs and Generative AI. A shocking 93-percent of machine learning engineers reported encountering issues with AI-based production models either on a daily or weekly basis, while 89-percent of these professionals also acknowledge encoun-



AI hallucination: A glass half full or empty?

tering hallucinations within these AI systems. According to the survey findings, these AI distortions often materialise as factual inaccuracies, biases, or potentially harmful content, underscoring the critical importance of implementing robust monitoring and control mechanisms to mitigate such AI hallucination issues effectively.

Can AI hallucinations be detected and stopped?

University of Oxford researchers seem to have made significant progress in ensuring the reliability of information generated by AI, one that addresses the issue of AI hallucination fair and square. Their study, published in Nature, introduces a novel method for detecting instances when LLMs hallucinate by inventing plausible-sounding but imaginary facts. The new method proposed by Oxford researchers analyses the statistics behind any given AI model's answer, specifically looking at the uncertainty in the meaning of a phrase in a generated sentence rather than just its grammatical structure, allowing it to determine if the model is genuinely unsure about the answer it generates for any given prompt. According to the researchers, their new method outperformed existing ones in detecting incorrect answers in GenAI based LLMs, leading to more secure deployment of GenAI in contexts where errors can have serious consequences, such as legal or medical question-answering.

Microsoft also claims to tackle AI hallucinations through new tools as part of its Azure AI Studio suite for enterprise customers, according to a report by The Verge. Microsoft is able to detect AI hallucinations in GenAI-based deployments of its enterprise customers' apps by blocking malicious prompts that trick their customers' AI into deviating from its training data. It also analyses the AI model's response to check if it contains fabricated information and further assess potential vulnerabilities in the AI model itself. These features readily integrate with popular GenAI models like GPT-4 and Llama, according to Microsoft, giving its Azure cloud users more control in preventing unintended and potentially damaging AI outputs.

Other big tech players aren't sitting idle in the face of AI hallucinations. Beyond recognising the importance of high-quality training data for LLMs, Google Cloud Platform employs techniques like regularisation, which penalises GenAI models for making extreme predictions, preventing overfitting to the training data and generating potentially hallucinating outputs. Amazon uses a similar approach in its online cloud empire, with AWS (Amazon Web Services) also exploring approaches like Retrieval-Augmented Generation (RAG), which combines the LLM's text generation capabilities with a retrieval system that searches for relevant information and helps the LLM stay grounded in factual information while generating text and reduce the chances of AI hallucination.

Long story short, it appears there's no single solution for stopping AI hallucinations. With GenAI deployments across various industries still accelerating, the problem of AI hallucination remains an ongoing area of research for all major tech players and academia. **AI**

